

4 Statistical physics

The problems are roughly ordered to coincide with the lectures. Those indicated with ♡. Any labelled with ♣ are the hardest ones, which might only occur as a "sting in the tail" at the end of a long examination question.

Problem 4.1 East, West, North and South

An otherwise empty box contains 16 randomly moving atoms. Consider the box to be divided into 4 equal volumes by mathematical surfaces, i.e. imaginary partitions. Any individual atom is equally likely to occupy any of the 4 sub-volumes.

- (i) What is the mean number of atoms in one of the four sub-volumes?
- (ii) What is the standard deviation in this number? [1.732]
- (iii) What is the standard deviation in the number of atoms in the remaining 3/4 of the volume? [1.732]

Problem 4.2 Large and small partitions

An ideal gas of N molecules occupies a container of volume V_0 . There is a sub-volume $V < V_0$. What is the mean number of atoms in the sub-volume and the standard deviation in this number when

1. $V \ll V_0$?
2. $V \rightarrow V_0$?

Problem 4.3 Counting partitions

Three distinguishable atoms each have 3 energy levels, 0, ε and 2ε . How many ways are there of getting a total energy of 3ε ?

Problem 4.4 Almost saturated magnet

In the spin $\frac{1}{2}$ paramagnet system we considered in lectures, we mostly dealt with small numbers of magnetic moments. In a real system of volume a few cm^3 , the number is more like 10^{22} . Suppose a magnetic field is applied to a system of 10^{22} paramagnets, and the energy of the system is fixed such that only 10% of the moments are antiparallel to the magnetic field. What is now the entropy of the system? [0.055 J K^{-1}]

Hint: you will need to use the crudest version of Stirling's approximation,

$$N! \underset{N \rightarrow \infty}{\sim} (N/e)^N.$$